fiskaltrust Middleware for Compliance-as-a-Service

Concept for a Middleware which can be used to accomplish Compliance-as-a-Service for point-of-sale systems. These systems can be cloud native or local deployed or hybrid. Simplification and less dependencies for point-of-sale systems by providing full fiscalization compliance and seamless integration with fiskaltrust ecosystem, including monetarization via add-on products and subscriptions, is the primary target to reach.

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Document Handling

This Document is a central point of information for product management and is used to synchronize with the whole organization of fiskaltrust. It should give any fiskaltrust employee and any partner used to fiskaltrust daily operations and overview of a product or service from multiple perspectives. The perspectives may be some of the following:

- Marketing: How value is provided to users, who is the target audience for the product or service, …
- Sales: What are relevant channels and what are the requirements to be able explain and sell the product or service, …
- Technical: What background services are required, what level of majority is required to use the product or service, what need to be given on fiskaltrust side to be able to operate the product, what service levels within the fiskaltrust organization are required and promised, …
- Distribution: How is the product or service shipped, …
- Support: Where are the touchpoints to provide help, what are the boundaries for operations, what service levels are required and promised, …

- Business: What strategic impact has the product on the overarching organization strategy, what dependencies do exist withing the business, what dreads and opportunities can be identified related to compliance, legal and ISMS, …
- Finance: What are requirements for preparing the environment, purchasing infrastructure, return on investments, long term profitability, impacts on revenue stream, …
- Any other relevant perspectives

It is owned by the functional area Product Management and therefore stored in a central place at [https://fiskaltrust.sharepoint.com/:w:/r/sites/fa_productmanagement/Freigeg-ebene%20Dokumente/middleware/fiskaltrust%20Middleware%20for%20compliance-as-a-service.docx?d=w773c16b013a74bcdb21920d8ba176732&csf=1&web=1&e=IysUSa], what also provides a Version history and different sharing models within fiskaltrust. Outside of fiskaltrust it should only be shared as a PDF version if this is required. When it is shared outside fiskaltrust for content updates, there needs to be a defined touchpoint manager, who can merge back the changes from outside the organization on a manual path.

As product definition is a living document, there are some additional rules which need to be cared on.

- Use Change Tracking when you do changes on existing already approved content, to signal your changes to the owner.
- Use Comments to ask questions within existing context or to challenge existing content. Comments need to be resolved by the person who opened the comment. When a question or change request within a comment is answered by a reply or by a change, the person who opened the comment is in charge of closing the comment.
- Use color coding to provide additional information on the described sections. Color code should be applied to the heading and is then valid for the complete section below the heading. If there should be multiple color coding within headings, you need to break up the section.
    - White background color: These sections are final descriptions, final requirements and implemented in the product or service from all perspectives and are generally available in production.
    - Yellow background color: These sections are under discussion, before implementation starts and before final requirements are clear. Around these sections there should be a heavy discussion with comments and a high number of changes.
    - Turquoise background color: These sections are valid descriptions which are currently in implementation and partially available as Alpha or Beta features and most of the time available in Sandbox. There may be changes in description and changes in implementation before this goes out as a preview. If there are feature flags involved, this may be provided to friendly customers to get early feedback.
    - Green background color: These sections are final descriptions, where product features are marked with preview and available in production. The provided functionality should be stable and tested against multiple perspectives and edge cases to get then general available.

Goal of the European Middleware

Europe has a diverse and broad spread field of requirements for point-of-sale system requirements related to fiskalisation. fiskaltrust will extend the open-source middleware to be able to use all countries in Europe step by step and as many devices and services as possible, to lower dependencies between point-of-sale software and single fiscal printer implementations, government services, and other security devices, independent if hardware or cloud based. To lower the market entry barrier for Implementation Partners, fiskaltrust is abstracting all local requirements and provides a seamless ecosystem for integration of a business opportunity for add-on products and services for different Channel Partners.

Countries

Austria: in production
Germany: in production
France: in production
Italy: in production
Greece: in production
Spain: in production
Portugal: in sandbox
Belgium: planned
Denmark: planned
Poland: planned

Basic Middleware concept

Middleware should abstract processes and data over multiple markets/countries.
Therefore, it uses an object of type “Queue” which has the same outside behavior related to functions, and data and implements an abstract security mechanism. Beyond this security mechanism, all national requirements are implemented and mapped towards the overarching outside data structure.
Within the queue object, this uses an object of type “SCU” (signature creation unit) to create at least a signature, identifying the issuer of the receipt data, but also implements all pure local related requirements for fiscalization. In Italy there are three specific methods to reach compliance at the point-of-sale.
Consumers can get the receipt by paper, by QR-code-scan, by e-mail, whatever matches best. But if the Consumer asks for a receipt, he needs to get on, as barrier-less as possible. Depending on the specific method, receipts have some specific required items on them.

Abstract security mechanism

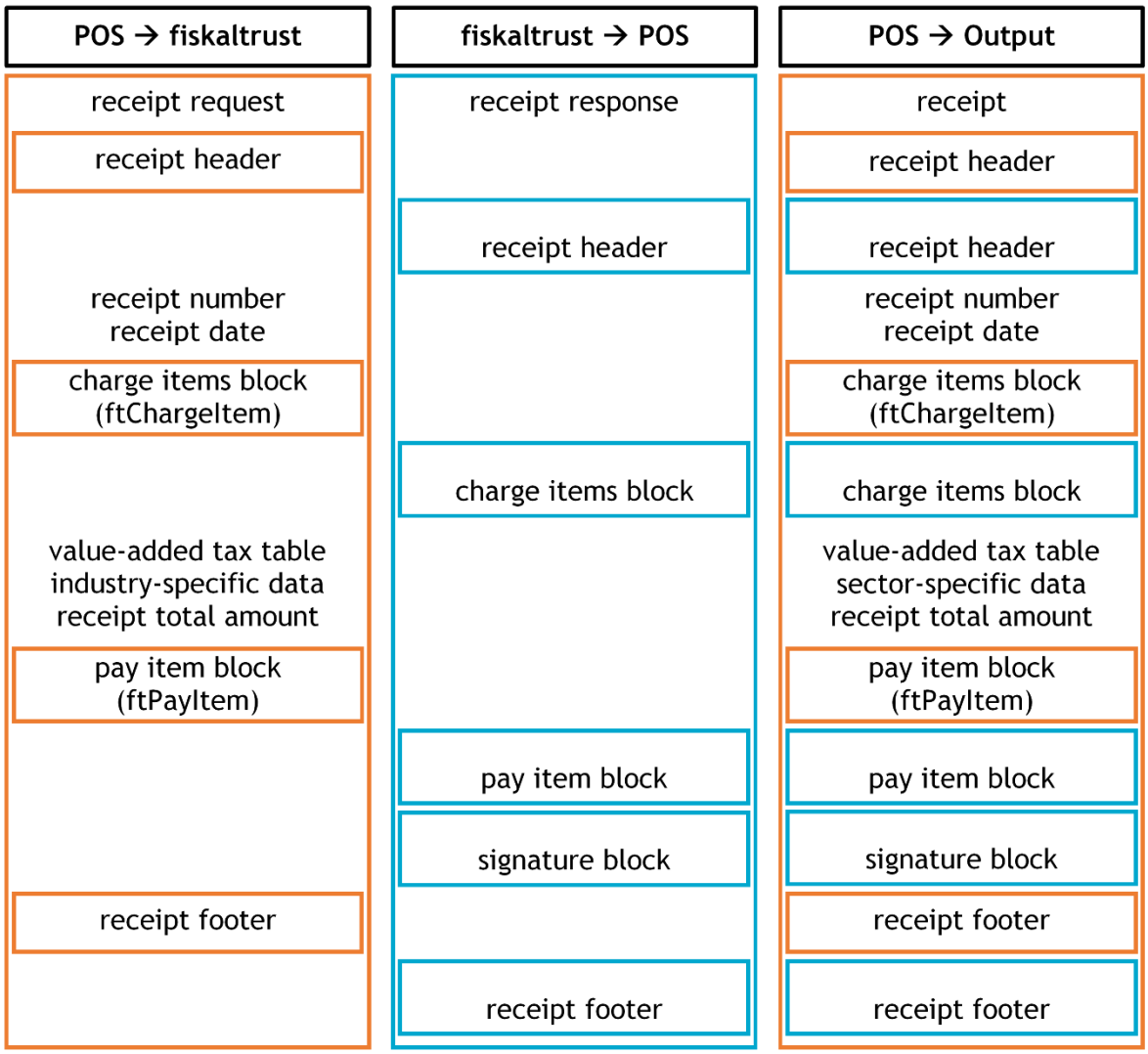
Instant and local temper prove mechanism is a hash-chain, which chains each successful request/response with the previous one. This hash-chain is often also called a receipt-chain.
[TBD picture receipt-chain]
To ensure the end of the chain is secured, an artefact of the data included within the hash-chain needs to be published as soon as possible. In many countries this is done by giving away a paper receipt. At fiskaltrust this is accomplished by mirroring the data towards a write only Cloud-API.
[TBD picture receipt data mirroring, helipad and/or digital receipt]

See also:

- Cash register integration | fiskaltrust Documentation Platform

Receipt generation

The method “sign” is called, and receipt details are provided via “ReceiptRequest”. After successfully processing the method, “ReceiptResponse” holds additional data for the receipt. The final receipt is generated by adding the sections of “ReceiptRequest” and “ReceiptResponse” to the final receipt.



Lifecycle management

Connectivity check [echo()]

This function provides fast and easy communication checks. The transferred message is sent back directly.

Initial Operation [sign(ReceiptCase=InitialOperationReceipt)]

Initializes a new fiskaltrust.Middleware instance. Depending on the fiscalisation device, this includes different actions related to the device.

Out-of-Operation [sign(ReceiptCase=OutOfOperationReceipt)]

Permanently disables the fiskaltrust.Middleware instance.

Daily operations

Issue Receipt [sign(ReceiptCase=CashReceipt|…|…)]

- POS receipt

Issue Invoice [sign(ReceiptCase=B2BInvoice|…|…)]

- Delivery note
- B2G Invoice
- B2B Invoice
- B2C Invoice
- Other/Order/Quote

Log CashChange [sign(ReceiptCase=PayTransfer|PayOut|…|…)

- Payment transfer receipt

Daily Closing [sign(ReceiptCase=DailyClosing|MonthlyClosing|YearlyClosing)]

- Shift closing
- Daily/Monthly/Yearly closing

Log Events [sign(ReceiptCase=Protocol|Log|…|…)

- Configuration change
- System restart

Special transaction/processing handling

ZeroReceipt [sign(ReceiptCase=ZeroReceipt|…|…)

- Statusinformation
- Communication issue resolution
-

Failure scenario

See also: [Failure scenarios | fiskaltrust Documentation Platform]

Message pending / Zero Receipt

When a message for the operator of the point-of-sale system is pending, all ReceiptResponses are flagged with the ftState 0x0040. The message can be claimed by issuing a ZeroReceipt by calling sign(ReceiptCase=ZeroReceipt) and the message is responded within the signatures.

Security Mechanism out of operation

When the local required security mechanism is not initialized or already out of operation, then all ReceiptResponses are flagged with the ftState 0x0001.

Audit

Journal

…

Logical Functions

PayItem = pay(PayRequest)

ChargeItem = order(OrderRequest)

ReceiptResponse = sign(ReceiptRequest)

IssueResponse = issue(ReceiptRequest, ReceiptResponse)

Datamodel

ReceiptRequest

description:	The cash register transfers the data of an entire receipt request to the fiskaltrust.Middleware using the ReceiptRequest data structure. The details of fields supported by this data structure are outlined in the table below. The field fiskaltrust receipt case (ftReceiptCase) is of the highest importance for the correct processing of the receipt. This field defines the receipt type, determines if the receipt has to be secured accordingly to national law, and establishes the way to calculate the correct values for each national counter.
cbTerminalID	**string** ***maxLength: 1023*** ***nullable: false*** ***default: undefined*** **The unique identification of the input-station/terminal within a cash-register/pos-system identified by ftCashBoxID.**
cbReceiptReference*	**string** ***maxLength: 1023*** ***nullable: false*** ***default: undefined*** **Reference number sent by the cash register. This value must be a unique string/receipt number related to the calling cash register. This string/receipt number is a unique primary key of the dataset of the cash register.**
cbReceiptMoment*	**string($date-time)** ***nullable: false*** ***default: now()*** ***example: 2020-06-29T17:45:40.505Z*** **moment at which the receipt was created by the cash register, must be provided in UTC.**
cbChargeItems*	**[** **list of line items related to services and products** **ChargeItem{...}]**
cbPayItems*	**[** **list of line items related to payments** **PayItem{...}]**

ftCashBoxID	**string($uuid)** ***nullable: false*** ***default: 00000000-0000-0000-0000-000000000000*** **identification of the cash register**
ftPosSystemID	**string($uuid)** ***nullable: false*** ***default: 00000000-0000-0000-0000-000000000000*** **identification of the used software of the cash register**
ftReceiptCase	**integer($uint64)** ***nullable: false*** ***default: 0*** **Type of business according to fiskaltrust reference. for detailed definition goto ftReceiptCase .this is relevant for fiskaltrust middleware processing and is a country specific mapping. as fallback when not specified, a cash register business case at the point-of-sale is used.**
ftReceiptCaseData	**{** **description:** optional gives additional details for defined type of business related to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO! **oneOf ->** **[...]** **{...}** ***nullable: false*** **}**
ftQueueID	**string($uuid)** ***nullable: true*** ***default: null*** **optional routing instruction to identify a specific queue behind a load balancer or in other usecases**
cbPreviousReceiptReference	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional Reference cbReceiptReference of the previous receipt. Used to connect multiple requests for a single Business Case.**
cbReceiptAmount	**{** **description:** optional total receipt amount, including value added taxes (=gross receipt amount) given to avoid calculation and rounding

		differences. systems which are using net amounts as central calculation should always use this property. if not provided, the sum of **Amount** in all provided **ftChargeItems** is used as total receipt amount. uses **DecimalPresissionMultiplier** as discriminator between int64 and double
	oneOf ->	**integer($int64)** **number($double)**
	}	
	nullable: true	
cbUser	**{**	
	description:	optional Identification of the user, who creates the receipt.
	oneOf ->	**[...]** **{...}** ***nullable: false***
	}	
cbArea	**{**	
	description:	optional Identification of the area/section/field, in which the receipt is created. e.g. Table number of a restaurant business; a department of a commercial establishment; the vehicle of a taxi company
	oneOf ->	**[...]** **{...}** ***nullable: false***
	}	
cbCustomer	**{**	
	description:	optional Identification of the consumer, for whom the receipt is created. e.g. Email address, phone number, personal tax number
	oneOf ->	**[...]** **{...}** ***nullable: false***
	}	
cbSettlement	**{**	
	description:	optional Settlement identification where this receipt will be added. e.g. shift number or day of operation
	oneOf ->	**[...]** **{...}** ***nullable: false***
	}	

Currency	**CurrencyEnumCurrencyEnumstring** ***nullable: false*** ***default: EUR*** **used as currency code for money numbers. along ISO 4217 (https://en.wikipedia.org/wiki/ISO_4217)** **Enum:** **[EUR, CHF, CZK, HUF, BAM, DKK, RON, NOK, PLN, RSD, SEK, UAH, USD, AED, AFN, ALL, AMD, ANG, AOA, ARS, AUD, AWG, AZN, BBD, BDT, BGN, BHD, BIF, BMD, BND, BOB, BOV, BRL, BSD, BTN, BWP, BYN, BZD, CAD, CDF, CHE, CHW, CLF, CLP, CNY, COP, COU, CRC, CUP, CVE, DJF, DOP, DZD, EGP, ERN, ETB, FJD, FKP, GBP, GEL, GHS, GIP, GMD, GNF, GTQ, GYD, HKD, HNL, HTG, IDR, ILS, INR, IQD, IRR, ISK, JMD, JOD, JPY, KES, KGS, KHR, KMF, KPW, KRW, KWD, KYD, KZT, LAK, LBP, LKR, LRD, LSL, LYD, MAD, MDL, MGA, MKD, MMK, MNT, MOP, MRU, MUR, MVR, MWK, MXN, MXV, MYR, MZN, NAD, NGN, NIO, NPR, NZD, OMR, PAB, PEN, PGK, PHP, PKR, PYG, QAR, RUB, RWF, SAR, SBD, SCR, SDG, SGD, SHP, SLE, SLL, SOS, SRD, SSP, STN, SVC, SYP, SZL, THB, TJS, TMT, TND, TOP, TRY, TTD, TWD, TZS, UGX, USN, UYI, UYU, UYW, UZS, VED, VES, VND, VUV, WST, XAF, XAG, XAU, XBA, XBB, XBC, XBD, XCD, XDR, XOF, XPD, XPF, XPT, XSU, XTS, XUA, XXX, YER, ZAR, ZMW, ZWL]**
DecimalPrecisionMultiplier	**DecimalPrecisionMultiplierEnumDecimalPrecisionMultiplierEnuminteger($int32)** ***nullable: false*** ***default: 1*** **is used as a multiplier for decimal numbers. When value 1 is used, relevant numbers are interpreted as floating-point numbers. for all other values, relevant numbers are interpreted as integer and need to be divided by the Multiplier to get the decimal representation.** **Enum:** **[1, 100, 10000, 1000000, 100000000]**

ReceiptResponse

description:	fiskaltrust.Middleware sends back the processed data to the cash register through the receipt response. The data included in the request, such as header, service, pay items, and footer, will not be sent back. The returned data is added to the receipt as supplement to the data of the receipt request.
ftQueueID	**string($uuid)** ***nullable: false*** ***default: 00000000-0000-0000-0000-000000000000*** **identification of the queue used for processing**
ftQueueItemID	**string($uuid)** ***nullable: false*** ***default: 00000000-0000-0000-0000-000000000000*** **identification of the item within a specific queue used for processing**
ftQueueRow*	**integer($uint64)** ***nullable: false*** ***default: 0*** **row in which the item is stored within a specific queue used for processing**
ftCashBoxIdentification*	**string** ***maxLength: 1023*** ***nullable: false*** ***default: undefined*** **human readable identification or serial number of the cash register related to the national regulation/law. This needs to be printed on the receipt to identify the cash register within a merchant and is unique over a single merchant. do not mix up ftCashBoxId with ftCashBoxIdentification. the ftCashBoxId identifies a configuration container and is used as part of authentication. The ftCashBoxIdentification is a human readable identification of the queue.**

ftReceiptIdentification*	**string** ***maxLength: 1023*** ***nullable: false*** ***default: ft0#*** **human readable identification of the receipt related to the national regulation/law and the ftCashBoxIdentification/ Queue. this need to be printed on the receipt to identify it within a merchant and cash register. this always starts with 'ft', followed by the row number of the queue in hexadecimal, followed by '#', followed by national required or defined receipt numbering.**
ftReceiptMoment*	**string($date-time)** ***nullable: false*** ***default: now()*** ***example: 2020-06-29T17:45:40.505Z*** **The moment at which the receipt was processed by fiskaltrust.Middleware, must be provided in UTC. This needs to be printed on the receipt at local date/time.**
ftSignatures*	**[** **list of signature items generated by fiskaltrust.Middleware. This needs to be printed on the receipt according to given format instructions to comply with national regulation/law and to enable fiskaltrust's compliance-as-a-service.** **SignatureItem{...}]**
ftState*	**integer($uint64)** ***nullable: false*** ***default: 0*** **indicates the status of the fiskaltrust.Middleware according to fiskaltrust reference. for detailed definition goto ftState .**
ftStateData	**{** **description:** optional gives additional details for the status of fiskaltrust.Middleware defined type of business related to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO! **oneOf ->** **[...]** **{...}** ***nullable: false*** **}**

ftCashBoxID	**string($uuid)** ***nullable: false*** ***default: 00000000-0000-0000-0000-000000000000*** **mirror from ReceiptRequest identification of the cash register**
cbTerminalID	**string** ***maxLength: 1023*** ***nullable: false*** ***default: undefined*** **mirror from ReceiptRequest The unique identification of the input-station/terminal within a cash-register/pos-system identified by ftCashBoxID.**
cbReceiptReference	**string** ***maxLength: 1023*** ***nullable: false*** ***default: undefined*** **mirror from ReceiptRequest Reference number sent by the cash register. This value must be a unique string/receipt number related to the calling cash register. This string/receipt number is a unique primary key of the dataset of the cash register.**
ftReceiptHeader	**[** **additional header lines that need to be printed on the receipt.** **string** ***maxLength: 1023*** ***nullable: false*]**
ftChargeItems	**[** **List of line items added by fiskaltrust.Middleware while processing the request related to services and products this need to be printed on the receipt.** **ChargeItem{...}]**
ftChargeLines	**[** **additional text for line items related to services and products this need to be printed on the receipt.** **string** ***maxLength: 1023*** ***nullable: false*]**

ftPayItems	**[** **list of line items added by fiskaltrust.Middleware while processing the request related to payments this need to be printed on the receipt.** **PayItem{...}]**
ftPayLines	**[** **additional text for line items related to payments this need to be printed on the receipt.** **[...]]**
ftReceiptFooter	**[** **additional footer lines that must be printed on the receipt.** **[...]]**

ChargeItem

description:	represents an item related to a service or a product, taxable
ftChargeItemID	**string($uuid)** ***nullable: true*** ***readOnly: true*** ***default: null*** **optional used as an identifier of a chargeitem, when reading data**
Quantity*	**{** **description:** defines the quantity of the line item. quantities of line items with same **Description**, **VATRate** and itemprice (=**Amount**/**Quantity**) can be accumulated for better visualization. uses **DecimalPresissionMultiplier** as discriminator between int64 and double **oneOf ->** **[...]** **[...]** **}**
Description*	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **defines the description of the line item. line items with same Description, VATRate and itemprice (=Amount/Quantity) can be accumulated for better visualization**
Amount*	**{** **description:** defines the (total) amount of the line item. to get itemprice the amount need to be devided by quantity. uses **DecimalPresissionMultiplier** as discriminator between int64 and double **oneOf ->** **[...]** **[...]** **}**
VATRate*	**{** **description:** defines the value added tax rate as a percentage of the line item. line items

		with same **Description**, **VATRate** and itemprice (=**Amount**/**Quantity**) can be accumulated for better visualization. uses **DecimalPresissionMultiplier** as discriminator between int64 and double
	oneOf ->	**[...]** **[...]**
	}	
ftChargeItemCase	**integer($uint64)** ***nullable: false*** ***default: 0*** **optional defines the type of service or product related to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO! This is relevant for fiskaltrust middleware processing and is a country specific mapping. As fallback when not specified, a service or product delivered at the point-of-sale with defined VATRate is used.**	
ftChargeItemCaseData	**{**	
	description:	optional gives additional details for defined type of service or product related to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO!
	oneOf ->	**[...]** **{...}** ***nullable: false***
	}	
VATAmount	**{**	
	description:	optional when given and not null this amount is used as the (total) value added taxes amount for the line item to avoid rounding when accumulating value added taxes. systems which are using net amounts as central calculation should alsways use this property. uses **DecimalPresissionMultiplier** as discriminator between int64 and double
	oneOf ->	**[...]** **[...]**

	} ***nullable: true***
Moment	**string($date-time)** ***nullable: true*** ***default: null*** ***example: 2020-06-29T17:45:40.505Z*** **optional moment at which the serivce or product was ordered or delivered, must be provided in UTC. accumulated line items get the minimum (first) moment. if not given, the cbReceiptMoment is used as fallback**
Position	**{** **description:** optional used to sort and group the line items for receipt visualization. accumulated line items get the minimum (first) position. uses **DecimalPresissionMultiplier** as discriminator between int64 and double with a specific instruction/flag in **ftReceiptCase** a grouping of multiple line items can be activated. if this is the case, **Position** is threaded as a decimal number, the whole number represents the grouped line item and the fraction part is used within the group **oneOf ->** **[...]** **[...]** **}**
AccountNumber	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional account number for bookkeeping export purpose**
CostCenter	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional cost center for cost accounting purpose**
ProductGroup	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null***

	optional product group related to line item
ProductNumber	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional product number related to line item**
ProductBarcode	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional product barcode related to line item**
Unit	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional unit of measurement for the line item. e.g. you have on one charging session of an electric vehicle, this would be Quantity 1 and the total amount of the session within amount. the unit of measurement could be kW for dc charging or min for ac charging.**
UnitQuantity	**{** **description:** optional the quantity related to the unit of measurement defined in **Unit**. uses **DecimalPresissionMultiplier** as discriminator between int64 and double e.g. you have on one charging session of an electric vehicle, this would be **Quantity** 1 and the total amount of the session within **amount**. if the unit of measurement is **kW** for dc charging the **UnitQuantity** could be 65.4 as an example. which would mean the line item represents a charging session with a total amout of power of 65.4kW. **oneOf ->** **[...]** **[...]** **}** ***nullable: true***
UnitPrice	**{**

	description: optional the price related to the unit of measurement defined in **Unit**. uses **DecimalPresissionMultiplier** as discriminator between int64 and double e.g. you have on one charging session of an electric vehicle, this would be **Quantity** 1 and as example the total amount of 30.7 of the session within **amount**. if the unit of measurement is **kW** for dc charging the **UnitQuantity** could be 65.4 as an example and for the given total amount the **UnitPrice** would be 0.5. which would mean the line item represents a charging session with a total amout of power of 65.4kW with a price per kW of 0.5. **oneOf ->** **[...]** **[...]** **}** ***nullable: true***
Currency	**CurrencyEnumCurrencyEnumstring** ***nullable: false*** ***default: EUR*** **used as currency code for money numbers. along ISO 4217 (https://en.wikipedia.org/wiki/ISO_4217)** **Enum:** **[EUR, CHF, CZK, HUF, BAM, DKK, RON, NOK, PLN, RSD, SEK, UAH, USD, AED, AFN, ALL, AMD, ANG, AOA, ARS, AUD, AWG, AZN, BBD, BDT, BGN, BHD, BIF, BMD, BND, BOB, BOV, BRL, BSD, BTN, BWP, BYN, BZD, CAD, CDF, CHE, CHW, CLF, CLP, CNY, COP, COU, CRC, CUP, CVE, DJF, DOP, DZD, EGP, ERN, ETB, FJD, FKP, GBP, GEL, GHS, GIP, GMD, GNF, GTQ, GYD, HKD, HNL, HTG, IDR, ILS, INR, IQD, IRR, ISK, JMD, JOD, JPY, KES, KGS, KHR, KMF, KPW, KRW, KWD, KYD, KZT, LAK, LBP, LKR, LRD, LSL, LYD, MAD, MDL, MGA, MKD, MMK, MNT, MOP, MRU, MUR, MVR, MWK, MXN, MXV, MYR, MZN, NAD, NGN, NIO, NPR, NZD, OMR, PAB, PEN, PGK, PHP, PKR, PYG, QAR, RUB, RWF, SAR, SBD, SCR, SDG, SGD, SHP, SLE, SLL, SOS, SRD, SSP, STN, SVC, SYP, SZL, THB, TJS, TMT, TND, TOP, TRY, TTD, TWD, TZS, UGX, USN, UYI, UYU, UYW, UZS, VED, VES, VND, VUV, WST, XAF, XAG,**

	XAU, XBA, XBB, XBC, XBD, XCD, XDR, XOF, XPD, XPF, XPT, XSU, XTS, XUA, XXX, YER, ZAR, ZMW, ZWL]
DecimalPrecisionMul-tiplier	**DecimalPrecisionMultiplierEnumDecimalPrecisionMulti-plierEnuminteger($int32)** ***nullable: false*** ***default: 1*** **is used as multiplier for decimal numbers. when value 1 is used, revlevant numbers are interpreted as floating-point numbers. for all other values, relevant numbers are inter-preted as integer and need to be devided by the Multiplier to get the decimal representation.** **Enum:** **[1, 100, 10000, 1000000, 100000000]**

PayItem

description:	represents a item related to a payment
ftPayItemID	**string($uuid)** ***nullable: true*** ***readOnly: true*** ***default: null*** **optional used as an identifier of a chargeitem, when reading data**
Quantity	**{** **description:** defines the quantity of the line item. quantities of line items with same **Description** and itemprice (=**Amount**/**Quantity**) can be accumulated for better visualization. uses **DecimalPresissionMultiplier** as discriminator between int64 and double **oneOf ->** **[...]** **[...]** **}**
Description*	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **defines the description of the line item. line items with same Description and itemprice (=Amount/Quantity) can be accumulated for better visualization**
Amount*	**{** **description:** defines the (total) amount of the line item. uses **DecimalPresissionMultiplier** as discriminator between int64 and double **oneOf ->** **[...]** **[...]** **}**
ftPayItemCase	**integer($uint64)** ***nullable: false*** ***default: 0*** **optional defines the type of payment related to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO! this is relevant for**

	fiskaltrust middleware processing and is a country specific mapping. as fallback when not specified, a cah payment at the point-of-sale is used.
ftPayItemCaseData	**{** **description:** optional gives additional details for defined type of payment related to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO! **oneOf ->** **[...]** **{...}** ***nullable: false*** **}**
Moment	**string($date-time)** ***nullable: true*** ***default: null*** ***example: 2020-06-29T17:45:40.505Z*** **optional moment at which the payment was executed, must be provided in UTC. accumulated line items get the minimum (first) moment. if not given, the cbReceiptMoment is used as fallback**
Position	**{** **description:** optional used to sort and group the line items for receipt visualization. accumulated line items get the minimum (first) position. uses **DecimalPresissionMultiplier** as discriminator between int64 and double with a specific instruction/flag in **ftReceiptCase** a grouping of multiple line items can be activated. if this is the case, **Position** is threaded as a decimal number, the whole number represents the grouped line item and the fraction part is used within the group **oneOf ->** **[...]** **[...]** **}**
AccountNumber	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional account number for bookkeeping export purpose**

CostCenter	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional cost center for cost accounting purpose**
MoneyGroup	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional group related to line item**
MoneyNumber	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional number related to line item**
MoneyBarcode	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional barcode or serialnumber related to line item**
Currency	**CurrencyEnumCurrencyEnumstring** ***nullable: false*** ***default: EUR*** **used as currency code for money numbers. along ISO 4217 (https://en.wikipedia.org/wiki/ISO_4217)** **Enum:** **[EUR, CHF, CZK, HUF, BAM, DKK, RON, NOK, PLN, RSD, SEK, UAH, USD, AED, AFN, ALL, AMD, ANG, AOA, ARS, AUD, AWG, AZN, BBD, BDT, BGN, BHD, BIF, BMD, BND, BOB, BOV, BRL, BSD, BTN, BWP, BYN, BZD, CAD, CDF, CHE, CHW, CLF, CLP, CNY, COP, COU, CRC, CUP, CVE, DJF, DOP, DZD, EGP, ERN, ETB, FJD, FKP, GBP, GEL, GHS, GIP, GMD, GNF, GTQ, GYD, HKD, HNL, HTG, IDR, ILS, INR, IQD, IRR, ISK, JMD, JOD, JPY, KES, KGS, KHR, KMF, KPW, KRW, KWD, KYD, KZT, LAK, LBP, LKR, LRD, LSL, LYD, MAD, MDL, MGA, MKD, MMK, MNT, MOP, MRU, MUR, MVR, MWK, MXN, MXV, MYR, MZN, NAD, NGN, NIO, NPR, NZD, OMR, PAB, PEN, PGK, PHP, PKR, PYG, QAR, RUB, RWF, SAR, SBD, SCR, SDG, SGD, SHP, SLE, SLL, SOS, SRD, SSP, STN, SVC, SYP, SZL, THB, TJS, TMT,**

	TND, TOP, TRY, TTD, TWD, TZS, UGX, USN, UYI, UYU, UYW, UZS, VED, VES, VND, VUV, WST, XAF, XAG, XAU, XBA, XBB, XBC, XBD, XCD, XDR, XOF, XPD, XPF, XPT, XSU, XTS, XUA, XXX, YER, ZAR, ZMW, ZWL]
DecimalPrecisionMultiplier	**DecimalPrecisionMultiplierEnumDecimalPrecisionMultiplierEnuminteger($int32)** ***nullable: false*** ***default: 1*** **is used as multiplier for decimal numbers. when value 1 is used, revlevant numbers are interpreted as floating-point numbers. for all other values, relevant numbers are interpreted as integer and need to be devided by the Multiplier to get the decimal representation.** **Enum:** **[1, 100, 10000, 1000000, 100000000]**

SignatureItem

description:	The signature of the receipt must comply with the national law. The signature data returned in the response must be visualized on the receipt related to format instruction and to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO! The signature entries can also be used to visualize hints and messages related to the fiskaltrust.SecurityMechanism.
ftSignatureItemID	**string($uuid)** ***nullable: true*** ***readOnly: true*** ***default: null*** **optional used as an identifier of a signautreitem, when reading data**
ftSignatureFormat*	**integer($uint64)** ***nullable: false*** ***default: 0*** **Format for displaying signature data according to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO!**
ftSignatureType*	**integer($uint64)** ***default: 0*** **Type of signature according to fiskaltrust reference. for detailed definition goto https://docs.fiskaltrust.eu !TODO!**
Caption	**string** ***maxLength: 1023*** ***nullable: true*** ***default: null*** **optional Heading, which has to be displayed as text above the signature data.**
Data*	**{** **description:** optional Signature content which has to be displayed in the specified format. **oneOf ->** **[...]** **{...}** ***nullable: false*** **}**

Reference Tables

As the Middleware abstracts processes and data over multiple markets/countries, there is a specific mapping for Italian market. This mapping is based upon the overall tagging system which gives the additional benefit of giving all receipts, chargeitems and payitems also a semantical value. The following section describes the overall format very quick.

Format

Every case that is sent to the middleware, or every Item that is being returned from the middleware is based upon this tagging system. For the tagging system we are using hex based numbers since they make things like, flagging and having a consistend numbering scheme easier.
The overall format is built up of 4 sections.

```
_4954_2000_0010_0001
_CCCC_vIII_gggg_xxxx
```

- CCCC (e.g 4954): ASCII of two letter ISO country code (https://en.wikipedia.org/wiki/ISO_3166-1) (e.g. IT = 4954)
- vIII (e.g. 2000): This section is for versioning the tagging system (currently v2) and for future use
- gggg (e.g. 0010): These items are used for flags. Flags can change the basic behavior of a given type, but will live the overall semantical meaning of a type the same. (e.g. voiding of a receipt)
- xxxx (e.g. 0001): The last category is usually case specific but always consists of 4 numbers

ReceiptRequest related mapping

ftReceiptCase

Format

CCCC_vlll_gggg_txcc

t … ReceiptCaseType

Type	Category	Description	
0	Receipt	A basic receipt that is generated as part of a POS sale. A receipt usually serves as proof of payment. The receipt is used after the transaction is done (if goods are received). This is the usual process that is done at a POS.	
1	Invoice	An invoice is generated for those cases where payment isn't handled immediately.	
2	DailyOperations	This category contains receipt cases that the Middleware requires for various downstream processes (e.g. book keeping)	
3	Log	Logs can be used for storing / securing events that need are needed for additional processing or downstream processes. (e.g. log for cash drawer opened)	
4	Lifecycle	These operations are used for changing the overall state of the Middleware. Depending on the local regulations these receipts are handed over as part of a notification (e.g. FinanzOnline)	

txcc … ReceiptCase

0000	Unknown receipt type Use this as a fallback, it is handled in the same way like a point-of-sale receipt	
0001	Point-Of-Sale receipt type	
0002	Payment transfer receipt type	
0003	Point-Of-Sale receipt without fiskalization obligation or with exeption on fiskalization regulation	
0004	E-Commerce receipt type	
0005	Delivery Note	
0006	Table Check [PT] / Note [FR] / Zwischenrechnung [AT] / ???	
0007	Proforma Invoice	
1000	Unknown invoice type (=> simplified invoice in the future)	
1001	B2C invoice type	
1002	B2B invoice type	
1003	B2G invoice type	
2000	Zero Receipt	
2001	(reserved) One Receipt	
2010	Shift Closing Receipt	
2011	Daily Closing Receipt	
2012	Monthly Closing Receipt	
2013	Yearly Closing Receipt	
3000	Protocol (unspecified type)	
3001	Protocol (technical event)	
3002	Protocol (audit event / accounting event)	
3003	Internal usage / Material consumption	
3004	Order (log order)	
3005	Pay (log payment)	
3010	Copy Receipt / Print existing Receipt	
3011	Archive Receipt / Archive all data Receipt	
marketspecific		
4001	Queue-Start-Receipt (Initial operations receipt)	
4002	Queue-Stop-Receipt (Out of operations receipt)	
4011	Initiate SCU-switch	
4012	Finish SCU-switch	
4021	Initiate migration	
4022	Finish migration	

gggg … global tagging/flag

ftReceiptCaseFlags

0001	Process as Late Signing Receipt The cash register lost connection to the queue and processed receipts without communicating with the queue. All processed receipts marked with the hint "Security mechanism not reachable" need to be sent to the queue with this maker.
0002	Training Receipt
0800	Group by Position-Number Position fields are threaded as a decimal number, , the whole number represents the grouped line item and the fraction part is used within the group. The sum of all ChargeItems within a position must count toward the total receipt amount. If the quantity and amount are 0,00, the quantity and amount will not be visualized for this line on the digital receipt. Independent if main or subitem.
8000	ReceiptRequest If you don't receive a response, try this flag first before taking any other action. This will return a stored result for example in case of a timeout when cashregister calls queue.

PosReceipt / Invoice only

0004	IsVoid Marks Receipt as Void to previous one. Mark lineitems also as IsVoid to signal clear data.
0008	Process as Handwritten Receipt During a power outage, the Cash register will not work, and the merchant hands out handwritten receipts. These handwritten receipts need to be sent to the Security Mechanism by using this flag.
0010	IssurIsSmallBusiness Businesses below a country-specific size in revenue need not declare VAT. With this marker, the receipt shows no VAT, all prices are gross, and a country-specific hint must be printed.
0020	ReceiverIsBusiness Specific data need to be placed onto the receipt.
0040	ReceiverIsKnown Characteristics related to UStG are given. For example, Name, Adress, VAT-ID, other local things.
0080	IsSaleInForeignCountry
0100	IsReturn/IsRefund Marks Receipt as Return of good or service.
0200	InvoiceProcessingOnly | InvoiceDelivery Is used if a queueitemid should be generated to use in later processing, eg.g. issue.
0400	HasTransportInformation If used, then transport information are included within the document.

ZeroReceipt / Dailyoperation only

0010	Request additional information from SCU TseInfo in Germany RTInfo in Italy SignatureCertificate in Austria, France
0020	Request Force Connection/Download from SCU
0040	Request Bypass Connection/Download from SCU
0080	Request Self-Test from SCU
0100	Request MasterData update
8000	Direct SCU communication. ftReceiptCaseData carry the request payload, within "eu.fiskaltrust.Middleware.SCU.[CC].[implementation]" and ftStateData carry the resonse payload, within the same proper-tyname.

LifecycleReceipt only

0010	Queue registration / de-registration only Bypass SCU factory default initialization
0020	Bypass SCU communication, execute while communication failed towards SCU

lll … local flags

AT

cba … c=reserved ; b=reporting ; a = scu related

DE

All Receipt Type (xxxx)

001	Implicit mode, create StartTransaction implicit for each ReceiptType, no need to call Start-Transaction-Receipt

Log operation (3xxx)

010	Start-Transaction (with technical log type 3001)
020	Update-Transaction (with technical log type 3001)
030	Delta-Transaction (with technical log type 3001)
040	Fail-Transaction-Receipt (with technical log type 3001)

FR

cba … c=reserved ; b=reporting ; a = scu related

IT

cba … c=reserved ; b=reporting ; a = scu related

001	[RT-Printer|RT-Server|Government Service] not reachable Responded in case of a zero-receipt and other hard dependencies to the service. (TBD can this be replaced by Request Bypass Connection/Download from SCU)

ReceiptCaseData

Reference in case of “Void”, when cbReceiptReference cannot be used because of source-receipt is in differen queue/system.
{ RT-Device-Serialnumber, Z-Number, Document-Number, Document-Moment }

Reference in case of “InvoicePayment”
{ }

ftChargeItemCase

Format

_CCCC_vlll_gggg_NNSV_

v=2; version=2

V … VAT

https://europa.eu/youreurope/business/taxation/vat/vat-rules-rates/index_en.htm

V (type of VAT)	Description	VAT table code	IT
0	Unknown VAT rate	G	
1	Discounted-1 VAT rate	B	
2	Discounted 2 VAT rate	C	
3	Normal VAT rate	A	
4	Super reduced 1 VAT rate	D	
5	Super reduced 2 VAT rate	E	
6	Parking VAT rate	F	
7	Zero VAT rate	H	
8	Not Taxable (in VAT context)	I	

S … Type of service

S (type of service)	Description
0	Unknown type of service
1	Delivery (supply of goods) (goods purchased and sold)
2	Other service (supply of service) (services are provided)
3	Tip For owner use V=0 to 7, related to total amount For Employee use V=8, Not Taxable
4	Voucher For Single-Use-Voucher use V=0 to 7 For Multi-Use-Voucher use V=8, Not Taxable Voucher Sale is a positive (+) amount. Voucher Redeem is a negative (-) amount. IsVoid can be applied to reverse amounts. Avoid to use this for Multi-Use-Voucher, use PayItem instead, with ShowInChargeItems flag. For Single-Use-Voucher, apply the ShowInPayItems flag to visualize it similar to payment and to keep the total amount unreduced.
5	Catalog service (service or good produced in house) ToBe-Modified!!!
6	Not own sales / Agency business
7	Own Consumption
8	Grant For Unreal Grant use V=0 to 7 For Real Grant use V=8
9	Receiveable Receiveable creation is negative (-) amount. Receiveable reduction is positive (+) amount. IsVoid can be applied to reverse amounts. Avoid to use this, use PayItem instead, with ShowInChargeItems flag.
A	Cash Transfer Cash Transfer to till is positive (+) amount. Cash Transfer from till is negative (-) amount. Only useable with V=8, Not Taxable. IsVoid can be applied to reverse amounts.
F	Super Specific type of tax Detailed definition in NN=nn V=8 required &&

NN (nature of VAT)	Description	IT	GR
00	usual VAT applies		
10	Not Taxable 1x can be used to specify more country specific details. For example, IGL	***NI (N3)** marker mandatory [10] not taxable - exports [11] non-taxable - intra-community supplies [12] non-taxable - transfers to San Marino [13] non-taxable - transactions assimilated to export supplies [14] non-taxable - following declarations of intent [15] non-taxable - other operations which do not contribute to the formation of the ceiling	*Marker to print [fiskaltrust] {myData} [11] (mydata:14) **Article 33**. non-taxable - intra-community supplies ***Χωρίς ΦΠΑ - άρθρο 33 του Κώδικα ΦΠΑ*** [12] (mydata:8) **Article 29**-Export of goods outside of **EU *Χωρίς ΦΠΑ - άρθρο 29 του Κώδικα ΦΠΑ*** [13] (mydata:28) TAXFREE retail to non eu citizens ***Χωρίς ΦΠΑ – άρθρο 29 περ. β' παρ.1 του Κώδικα ΦΠΑ*** [14] (mydata:16) **Article 45** Includes the special regime of payment of tax from the receiver of goods and services NOT THE ISSUER ***Χωρίς ΦΠΑ - άρθρο 45 του Κώδικα ΦΠΑ*** [15] (mydata:6) **Article 24** Bottle package recycling, sales of tickets, sales of newspapers and magazines ***Χωρίς ΦΠΑ - άρθρο 24 του Κώδικα ΦΠΑ***

			[16] (mydata:7) **article 27** Services in Greece for medical services, services provided from doctors, dentists, insurance services, bank services, sale of 1st residence ***Χωρίς ΦΠΑ - άρθρο 27 του Κώδικα ΦΠΑ***
20	Not Subject 2x can be used to specify more country specific details.	*NS (N2) marker mandatory [20] not subject to VAT pursuant to articles from 7 to 7-septies of Presidential Decree 633/72 [21] not subject, other cases	
30	Exempt 3x	*ES (N4) marker mandatory [30] exempt	[31] (mydata:20) **article 50** Comments Special regime for travel agencies, travel packages taxed in GR. ***ΦΠΑ εμπεριεχόμενος - άρθρο 50 του Κώδικα ΦΠΑ*** [32] (mydata:9) **article 30** It concerns the exemptions applied under special customs regimes. Goods placed in special customs regimes (e.g., customs warehousing, active processing, etc.) are exempt

			from VAT. ***Χωρίς ΦΠΑ - άρθρο 30 του Κώδικα ΦΠΑ*** [33] (mydata:15) **Article 44** It concerns the special regime for small businesses(below 10K invoices). ***Χωρίς ΦΠΑ - άρθρο 44 του Κώδικα ΦΠΑ***
40	Margin scheme Do not print/show VAT rate and amount on receipt/invoice. 4x can be used to specify more country specific details.	***RM (N5)** marker mandatory [40] margin scheme / VAT not shown on the invoice	[41] (mydata:22) **Article52** Special taxation regime for taxable resellers who deliver second-hand goods and objects of artistic, collector's or archaeological value ***Χωρίς ΦΠΑ εμπεριεχόμενος - άρθρο 52 του Κώδικα ΦΠΑ***
50	Reverse charge 5x	***AL (N6)** marker mandatory [50] reverse charge - disposal of scrap and other salvage materials [51] reverse charge - sale of gold and silver pursuant to law 7/2000 as well as used jewelery to OPO [52] reverse charge - subcontracting in the construction sector	[51] (mydata:19) **Article54** The delivery, intra-Community acquisition and import of investment gold, including investment gold for which there are certificates, by type or by type or which is the subject of a transaction between gold accounts, including, in particular, gold loans and

		[53] reverse charge - sale of buildings [54] reverse charge - supply of mobile phones [55] reverse charge - supply of electronic products [56] reverse charge - performance in the construction sector and related sectors [57] reverse charge - energy sector transactions [58] reverse charge - other cases	swaps, with a right of ownership or claim to investment gold, as well as investment gold transactions with futures and forward contracts, which cause a change of ownership or claim to investment gold, ***Χωρίς ΦΠΑ - άρθρο 54 του Κώδικα ΦΠΑ***
60	VAT paid in other EU country 6x	(N7) marker mandatory [60] paid in another EU country (provision of telecommunications, broadcasting and electronic services pursuant to art. 7-octies, paragraph 1 letter a, b, art. 74-sexies of Presidential Decree 633/72)	[61] (mydata:3) **article 17** Sales of goods which DURING SALES are located outside of Greece, sales on boats and or planes during an intra-eu sale ***Χωρίς ΦΠΑ - άρθρο 17 του Κώδικα ΦΠΑ*** [62] (mydata:4) **article 18** Services taxed outside fo Greece including restaurant and catering services provided abroad INCLUDING SERVICES provided digitally when the receiver is living abroad. ***Χωρίς ΦΠΑ - άρθρο 18 του Κώδικα ΦΠΑ***
70	VAT distribution	*VI (VI) is a fiscal VAT (IVA) regime	

		that certain retailers can adopt. It allows the global registration of the daily takings amount without distinguishing the individual VAT rates. It only ever applies to goods.	
80	Excluded 8x	*EE (N1) marker mandatory [80] excluded pursuant to art. 15 of Presidential Decree 633/72	[81] (mydata:1) **Article 2&3** Includes transactions outside the scope of VAT (e.g. compensations for material damages, income from participations, subsidies, grants, etc., as well as the special regime of Mount Athos. ***Χωρίς ΦΠΑ - άρθρο 2 και 3 του Κώδικα ΦΠΑ*** [82] (mydata:2) **Article 5** Case of transfer of assets of a business as a) a whole, b) a branch, or c) a part of it through onerous or gratuitous cause or in the form of contribution to an existing or newly established legal entity. ***Χωρίς ΦΠΑ - άρθρο 5 του Κώδικα ΦΠΑ*** [83] (mydata:10) **Article 31** Rare

			case of tax warehouses sales ***Χωρίς ΦΠΑ - άρθρο 31 του Κώδικα ΦΠΑ*** [84] (mydata:11) **Article 32** It includes exemptions applicable to: certain categories of ships and watercraft and aircraft, for diplomatic and consular authorities, recognized international organizations, the European Community, the European Central Bank, etc., NATO and its organizations, to meet the needs of refugees and vulnerable groups, public donors, etc. for certain transports for [85]] (mydata:12) Article 32.Open seas ships [86]] (mydata:13) Article 32.1.Open seas ships [87] Χωρίς ΦΠΑ - ΠΟΛ.1029/1995 [88] (mydata 26) special case where you don't pay vat as long as the goods or services

			are intented for another EU state or 3rd party country ***Χωρίς ΦΠΑ - άρθρο 32 του Κώδικα ΦΠΑ***

nn (nature of non-VAT / super specific tax) V==8 && S==F)	Description	IT	GR
00	Exact description is used for mapping and printing.		
10			
20			
30			
40			
50			
60			
70			
80			

lll … local tagging/flag

gggg … global tagging/flag

0001	IsVoid Marks ChargeItem as Void previous position. Quantity and amount are inverted, related to original item.
0002	IsReturn/IsRefund Marks ChargeItem as Return of good or service. Quantity and amount are inverted, related to original item.
0004	Discount Marks ChargeItem as Discount/Extra for previous position. Positive (+) amount is extra. Negative (-) amount is discount. IsVoid or IsReturn/IsRefund will invert this behavior.
0008	Downpayment Marks ChargeItem as a downpayment. Positive (+) amount is the creation of downpayment. Negative (-) amount is reduction of downpayment. IsVoid or IsReturn/IsRefund will invert this behavior.
0010	Returnable Marks ChargeItem as a returnable. Positive (+) amount/quantity is handout. Negative (-) amount/quantity is reverse. IsVoid or IsReturn/IsRefund will invert this behavior.
0020	TakeAway Marks ChargeItem as TakeAway item to prove special VAT application
4000	Respond in ReceiptResponse
8000	ShowInPayments Visualize the item after Total Amount. This inverts amount and does not include the amount into the visualized total amount on the receipt.

ftPayItemCase

Format

_CCCC_vlll_gggg_xxPP_

PP … payment type

	Description	GR
00	Unknown	
01	Cash	GR: Cash
02	NonCash	
03	Crossed cheque	GR: Cheque
04	Debit Card	GR: POS / e-POS
05	Credit Card	GR: POS / e-POS
06	Voucher	GR: Cheque
07	Online	
08	Loyalty program Customer card	
09	Accounts receivable	GR: On Credit
0A	SEPA transfer (wire transfer)	GR: Default maps to Domestic payments GR: **Description == "IRIS"** maps to **IRIS** GR: **Description == "RF code payment (Web banking)"** maps to **Web banking**
0B	Other Bank transfer	GR: maps to **International Business Payment Account**
0C	Transfer to Cashbook / Vault / Owner / Employee Positive (+) amount contributes to cashbook/vault. This highers the amount in cashbook/vault. Negative (-) amount lowers the amount in cashbook/vault.	
0D	Internal / Material consumption	
0E	Grant	
0F	Ticket Restaurant / (Sodexo, edenred, usw.)	

v=2; version=2

gggg … global tagging/flag

0001	IsVoid

	Marks PayItem as Void previous position. Quantity and amount are inverted, related to original item. **IsVoid is used in cases where the exchange of money has not been executed yet.**
0002	IsReturn/IsRefund Marks PayItem as Return of good or service. Quantity and amount are inverted, related to original item. **IsReturn/IsRefund is used in cases where the exchange of money has been executed already.**
0004	(reserved)
0008	Downpayment Marks PayItem as a downpayment. Positive (+) amount is reduction of downpayment. Negative (-) amount is creation of downpayment.
0010	IsForeignCurrency Amount is still in EUR, at the moment of acceptance. ftPayItemData requires two data elements with “foreignCurrencySymbol” and “foreignCurrencyAmount” to persist data for daily closing and later bookkeeping transactions
0020	IsChange Usually contains a negative (-) amount. (IsVoid => can be inverted)
0040	IsTip Must be a negative (-) amount to flow out of payment method. ShowInChargeItems flag can be used to raise the total amount by the tip amount, to have a more convenient visualization.
0080	IsDigital/IsElectronic Electronic money, digital money
0100	IsInterface/AmountVerified? Was verified by interface, automated amount transfer
4000	Respond in ReceiptResponse
8000	ShowInChargeItems Visualize the item before Total Amount. This inverts amount and does include the amount into the visualized total amount on the receipt.

ReceiptResponse related mapping

ftReceiptnumber

ft{ReiceiptNumeratorHex}#{RT-Device-Z-Number}-{RT-Device-recNumber}

ftState

Format

_CCCC_vlll_gggg_gggg_

v=2; version=2

gggg_gggg … global tagging/flag

0000_0001	Security Mechanism is out Out of Operation. Queue is not started or already stopped.
0000_0002	SCU (Signature Creation Unit) temporary out of service. For at least one receipt, it was not possible to receive the signature from an allocated SCU, therefore the security mechanism has been put into "signature creation device out of service" mode. Regardless of whether an allocated SCU is available again or not, the mode remains in place until a ZeroReceipt cleans up the state and takes the market specific action required.
0000_0008	Deferred Queue Mode / Late Signing Mode is active. When the cash register doesn't reach the queue, it queues up the receipt requests while continuing to do business. Also, with a major failure of the cash register or a power outage, handwritten paper receipts are queued up while continuing to do business. After getting back to a full functional state, these queued-up ReceiptRequests are sent to the queue, having the original cbReceiptMoment of the business case and ReceiptCase tagged/flagged with 0001 (Deferred Queue / Late Signing) or 0008 (Handwritten). A result of this is a marker within the ftState, which can be resolved via ZeroReceipt. The reason for the marker is a mismatch between processed time along the receipt chain and a manual event to clean up the state and maybe notify 3rd parties of an outage.
0000_0040	Message Pending Middleware/Queue is a headless background service, but there are situations where communication with the cashier/operator or the cash register is necessary. For example, if the last daily closing was missed or if a special condition related to the signature creation unit or service happened. This is the moment when the message pending flag is set by the middleware, which should be signalled to the cashier by the POS system. By executing a ZeroReceipt, the cashier can read the message or instruction on the printed or displayed receipt.

	Related to local regulations, this receipt may be stored/archived with/for bookkeeping purposes; if this is the case, this is also visualized.
0000_0100	DailyClosing due When the first cbReceiptMoment used since the last DailyClosing and the current/latest cbReceiptMoment in the ReceiptRequest have a date-gap of more than two days (for example, the first since the last daily closing is 24/08 and the current is 26/08), then this state indicates, a Daily Closing should be done. DailyClosing is an essential part of the security mechanism and also executes additional market-specific cleanup tasks. Therefore, each queue should do a DailyClsoing to clear persistent changes in business data and also changes in the business period.
0000_0200	MonthlyClosing due When the first cbReceiptMoment used since last MonthlyClosing and the current/latest cbReceiptMoment in the ReceiptRequest are different, then this state indicates, a MonthlyClosing should be done.
0000_0400	YearlyClosing due
EEEE_EEEE	Error. Something went wrong while processing the last request. QueueItem exists but didn't reach the state of a ReceiptItem and didn't consume a ftReceiptNumber within the chain. Error reason is shown within the responded ftSignatureItems. This happens, for example, if the ReceiptCase is not recognized or is wrong.
FFFF_FFFF	Fail. Something went wrong while processing the last request, and nothing persisted within the Queue. Fail reason is shown within the responded ftSignatureItems. This happens, for example, when the flag ReceiptRequest is used after a communication outage, and no properly processed item is found.

lll … local flags

AT

cba … c=reserved ; b=reporting ; a = scu related

001	SCU permanent out of service 48h FinanzOnline timeout reached.
002	Backup SCU in use

DE

cba … c=reserved ; b=reporting ; a = scu related

001	SCU in switching state The queue is in the process of switching SCUs. This state is returned in case any receipts are processed between the initialize-switch receipt and the finish-switch receipt. These receipts are protected by the fiskaltrust.SecurityMechanism, but not sent to any TSE, as no SCU is connected at this point.

FR

cba … c=reserved ; b=reporting ; a = scu related

IT

cba … c=reserved ; b=reporting ; a = scu related

001	[RT-Printer|RT-Server|Government Service] not reachable Responded in case of a zero-receipt and other hard dependencies to the service.

ES

cba … c=reserved ; b=reporting ; a = scu related

ftSignature

ftSignatureFormat

p_ffff_
ffff … format

0000	Unknown / no format defined
0001	Text
0002	Link
0003	QR-Code (2D Code)
0004	Code128 (Barcode)
0005	OCR-A (optical character recognition, possible for Base32 data)
0006	PDF417-(2D Code)
0007	DATAMATRIX-(2D Code)
0008	AZTEC-(2D Code)
0009	EAN-8 (Barcode)
000A	EAN-13 (Barcode)
000B	UPC-A (Barcode)
000C	Code39 (Barcode, possible for Base32 data)
000D	Base64 (Raw Data)

p … position
Basic Layout is
[Header]
[ChargeItemBlock]

[TotalTaxBlock]
[PayItemBlock]
[SignatureBlock]
[Footer]
SignatureItems should be printed/visualized at the Receipt within the SignatureBlock sorted as received according additional format and visualization flags. First line in text format the caption of the SigntureItem and by a new line the content of the SignatureItem according the format.

0	After PayItemBlock / Before Footer
1	After Header / Before ChargeItemBlock
2	After ChargeItemBlock / Before TotalTaxBlock
3	After TotalTaxBlock / Before PayItemBlock
4	After Footer
5	Before Header

0100	Print/Visualize after Header (usual position is after PayItems / before Footer)
0200	Print/Visualize after ChargeItems (usual position is after PayItems)
0400	Print/Visualize after Total (usual position is after PayItems)
0800	Print/Visualize after ChargeItems (usual position is after PayItems)

ftSignatureFormatFlags

ftSignatureType

_CCCC_vlll_gggg_tsss

v=2; version=2

t … Type/Category

0	Uncategorized, Normal use (notification)
1	Information (notification), low priority
2	Alert (notification), high priority
3	Failure (notification), high priority

sss … SignatureCase

Case	Description	Caption
000	Notification	
001	Primary market related compliance signature	

010??	Middleware-version => the version of the middlware being used to generate the given receipt	
Exx??	Related information to EEEE_EEEE ftState Flag=>do not print/visualize Data=>Base64 stack trace if debug/sandbox	Exception Number/Name
Fxx??	Related information to FFFF_FFFF ftState Flag=>do not print/visualize Data=>Base64 stack trace if debug/sandbox	Exception Number/Name

gggg … global flags

0001	Archiving required. Signatures marked with this flag are known to be archived related to market specific bookkeeping requirements. In case of offline usage or pure open-source usage, receipts/artefacts having this flag need to be handled as bookkeeping/accounting-relevant item.
0010	Printing/Visualization is optional
0020	Do not print/visualize
0040	Printed receipt only
0080	Digital receipt only

AT

sss … SignatureCase

Case	Description	Caption
001	Signature/Payload according to RKSV	[www.fiskaltrust.at]
002	Daily operation notification	
003	FinanzOnline notification 2D code to execute FinanzOnline notification in case of offline usage or pure open-source usage.	

DE

sss … SignatureCase

Case	Description	Caption
001	Signature/Payload according to Kassen-SichV [qr] format	[www.fiskaltrust.de]
010	Start-transaction-result	

	[do-not-print] flag	
011	Finish-transaction-payload [do-not-print] flag	
012	Finish-transaction-result [do-not-print] flag	
013	QR-Code Version for receipt [optional-print] flag	
014	Cashregister Serialnumber for receipt [optional-print] flag	
015	ProcessType for receipt [optional-print] flag	
016	ProcessData for receipt [optional-print] flag	
017	Transaction number for receipt [optional-print] flag	
018	Signature counter for receipt [optional-print] flag	
019	Transaction start time for receipt [optional-print] flag	
01A	Signature logtime for receipt [optional-print] flag	
01B	Signature algorithm for receipt [optional-print] flag	
01C	Signature logtime format for receipt [optional-print] flag	
01D	Signature for receipt [optional-print] flag	
01E	Public key for receipt [optional-print] flag	
01F	Process start time for receipt	
020	Update-transaction-payload [do-not-print] flag	
021	Update-transaction-result [do-not-print] flag	
022	Certification identification and protec-tion profile restrictions for receipt	
023	TSE serial number for receipt	

FR

sss … SignatureCase

Case	Description	Caption
001	Signature/Payload according to French security mechanism. (BOI-TVA-DECLA-30-10-30) This is a JWT	[www.fiskaltrust.fr]

	[qr] format	
010	Shift closing payload	
011	Day closing payload	
012	Month closing payload	
013	Year closing payload	
014	Archive Totals payload	
015	Perpetual Totals payload	
	Zertification status of cashregister ???	

IT

sss … SignatureCase

Case	Description	Caption
001	Print below [PayItemBlock] / Document fiscalization RT-Printer [DATE(DD-MM-YYYY)] [TIME(HH:MM)][NEWLINE] DOCUMENTO N. [z-Number]-[Document-Number][NEWLINE] Codice Lotteria: [LotteryID][NEWLINE] Codice Fiscale: [CustomerID][NEWLINE] RT [SERIALNUMBER][NEWLINE] RT-Server [DATE(DD-MM-YYYY)] [TIME(HH:MM)][NEWLINE] DOCUMENTO N. [z-Number]-[Document-Number][NEWLINE] Codice Lotteria: [LotteryID][NEWLINE] Codice Fiscale: [CustomerID][NEWLINE] Server RT [SERIALNUMBER][NEWLINE] Cassa [ftCashboxIdentification][NEWLINE] -----FIRMA ELETTRONICA-----[NEWLINE] [SHA-METADATA][NEWLINE] ----------[NEWLINE] FE (Fattura Elettronica) [DATE(DD-MM-YYYY)] [TIME(HH:MM)][NEWLINE] [VAT-ID][NEWLINE] [ftReceiptIdentification][NEWLINE] [ftCashboxIdentification][NEWLINE]	[www.fiskaltrust.it]

	If number/signature is present/not null	
002	Print below [Header] / Document Type Document-Type (1) Vendita ~~DOCUMENTO COMMERCIALE[NEW-LINE]~~ di vendita o prestazione[NEWLINE] Document-Type (3) Reso ~~DOCUMENTO COMMERCIALE[NEW-LINE]~~ emesso per ANNULLAMENTO[NEW-LINE] [NEWLINE] Documento di riferimento: N. [reference-z-Number]-[reference-Document-Number] del [reference-Document-Date(DD-MM-YYYY)][NEWLINE] N. [reference-z-Number]-[reference-Document-Number] del [reference-Document-Date(DD-MM-YYYY)][NEWLINE] RT [reference-RT-Printer-Serial-number][NEWLINE] Server RT [reference-RT-Server-Serial-number][NEWLINE] Cassa [reference-ftCashboxIdentification][NEWLINE] XXX(*) del [reference-Document-Date(DD-MM-YYYY)][NEWLINE] Document-Type (5) Annullo ~~DOCUMENTO COMMERCIALE[NEW-LINE]~~ emesso per RESO MERCE[NEWLINE] [NEWLINE] Documento di riferimento: N. [reference-z-Number]-[reference-Document-Number] del [reference-Document-Date(DD-MM-YYYY)][NEWLINE] N. [reference-z-Number]-[reference-Document-Number] del [reference-Document-Date(DD-MM-YYYY)][NEWLINE]RT	DOCUMENTO COMMERCIALE

	[reference-RT-Printer-Serialnum-ber][NEWLINE] Server RT [reference-RT-Server-Serial-number][NEWLINE] Cassa [reference-ftCashboxIdentifica-tion][NEWLINE] XXX(*) del [reference-Document-Date(DD-MM-YYYY)][NEWLINE] If number is on same system If number is on different system If number is unknown XXX(*) "POS" in the case of a POS receipt; "VR" in the case of returnable contain-ers; “ND” in other cases	
010	RT printer/server serial number	
011	RT printer/server Z-Number	
012	RT printer/server Document-Number	
013	RT printer/server Document-Moment	
014	RT printer/server Document Type	
015	RT printer/server LotteryID	
016	RT printer/server CustomerID	
017	RT server SHA Metadata	
018	Amount	
	ReceiptURL ???	
	ESC/POS content ???	
	???? "serialNumber": "99IEC018305", "recNumber": "18", "zRepNumber": "Z0007", "recDate": "2023-08-28T13:21:32+02:00", "recAmount": "5077.14", "customerId": "123456789 ", "lotteryId": " ", "docType": "SALES",	
020	RT Reference ZNumber	
021	RT Reference DocNumber	
022	RT Reference Document Moment	

ES

sss … SignatureCase

Case	Description	Caption

PT

sss … SignatureCase

Case	Description	Caption

GR

sss … SignatureCase

Case	Description	Caption

ftJournalType

Format

_CCCC_vlll_gggg_tjjj_
v=2; version=2

t … Type/Cathegory

0	Common
1	Market specific

jjj … JournalCase

Case	Description
000	Version Information
001	ActionJournal
002	ReceiptJournal
003	QueueItemJournal

gggg … global flags

0001	Use ZIP compressed stream

AT

jjj … JournalCase

Case	Description
000	Status Information QueueAT
001	RKSV-DEP-Export

DE

jjj … JournalCase

Case	Description
000	Status Information QueueDE
001	.TAR-File-Export passthrough from TSE device Limited to data at device. Usually, data are purged from device after successful export.
002	DSFinV-K Export ZIP compression required.
003	.TAR-File-Export

FR

jjj … JournalCase

Case	Description
000	Status Information QueueFR
001	Ticket ("T" group) export
002	Payment Prove ("P" group) export
003	Invoice ("I" group) export
004	Grand Total ("G" group) export
007	Bill ("B" group) export
008	Archive ("A" group) export
009	Log ("L" group) export
00A	Copy ("C" group) export
00B	Training ("X" group) export
010	Export (in conjunction with Archiv)

IT

jjj … JournalCase

Case	Description
000	Status Information QueueIT
001	TBD passthrough from RT device
002	
003	

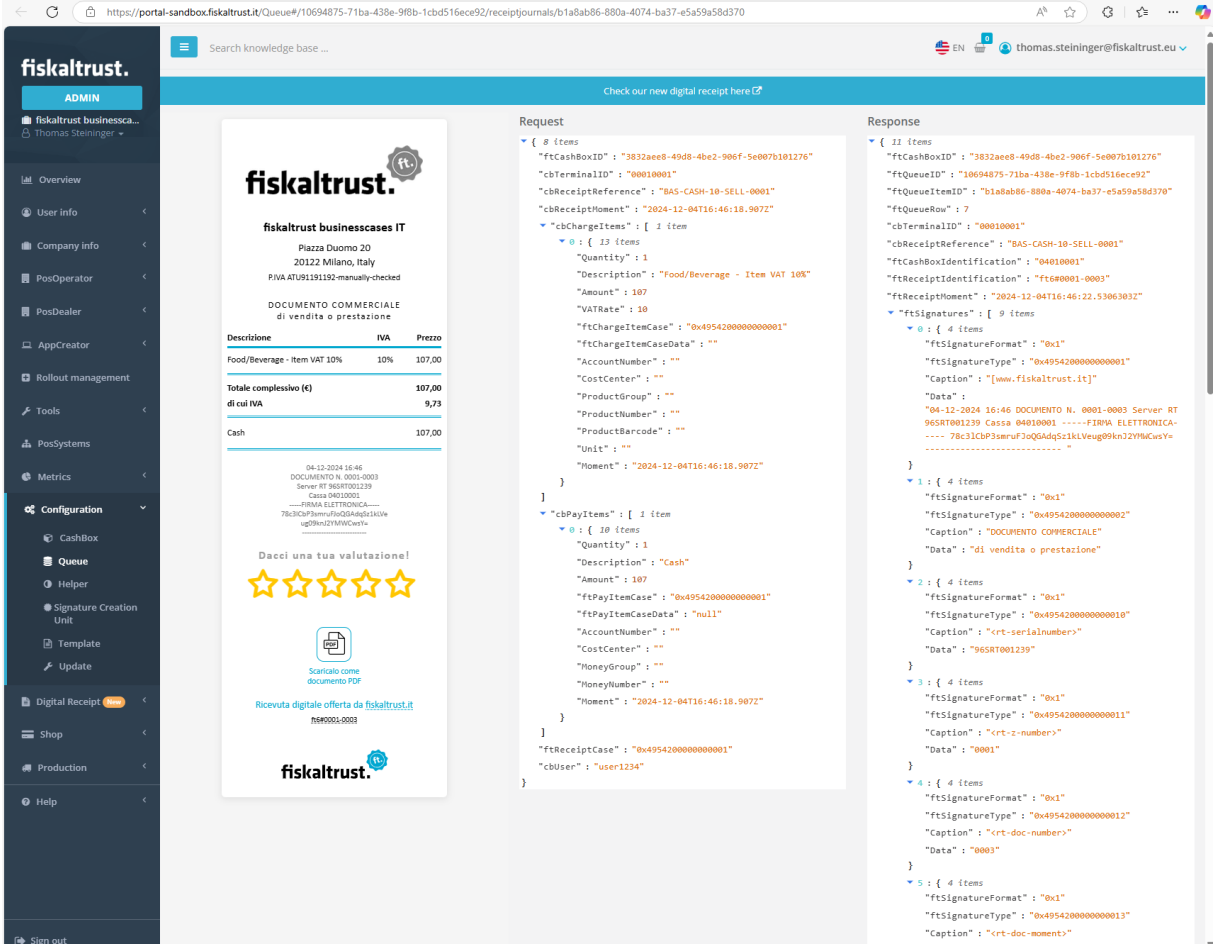
ES

jjj … JournalCase

Case	Description
000	Status Information QueueES
001	
002	
003	

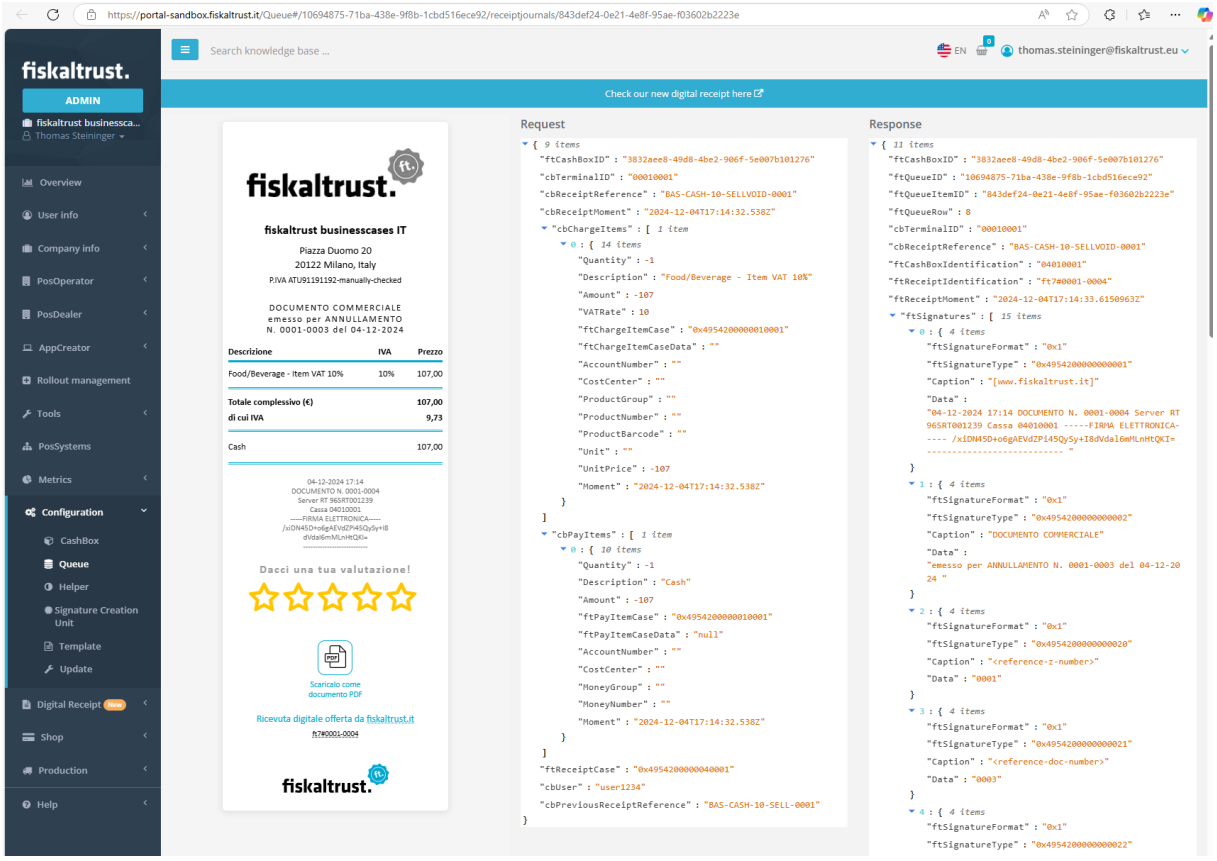
Common Usecase Examples

Cash sale
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/b1a8ab86-880a-4074-ba37-e5a59a58d370



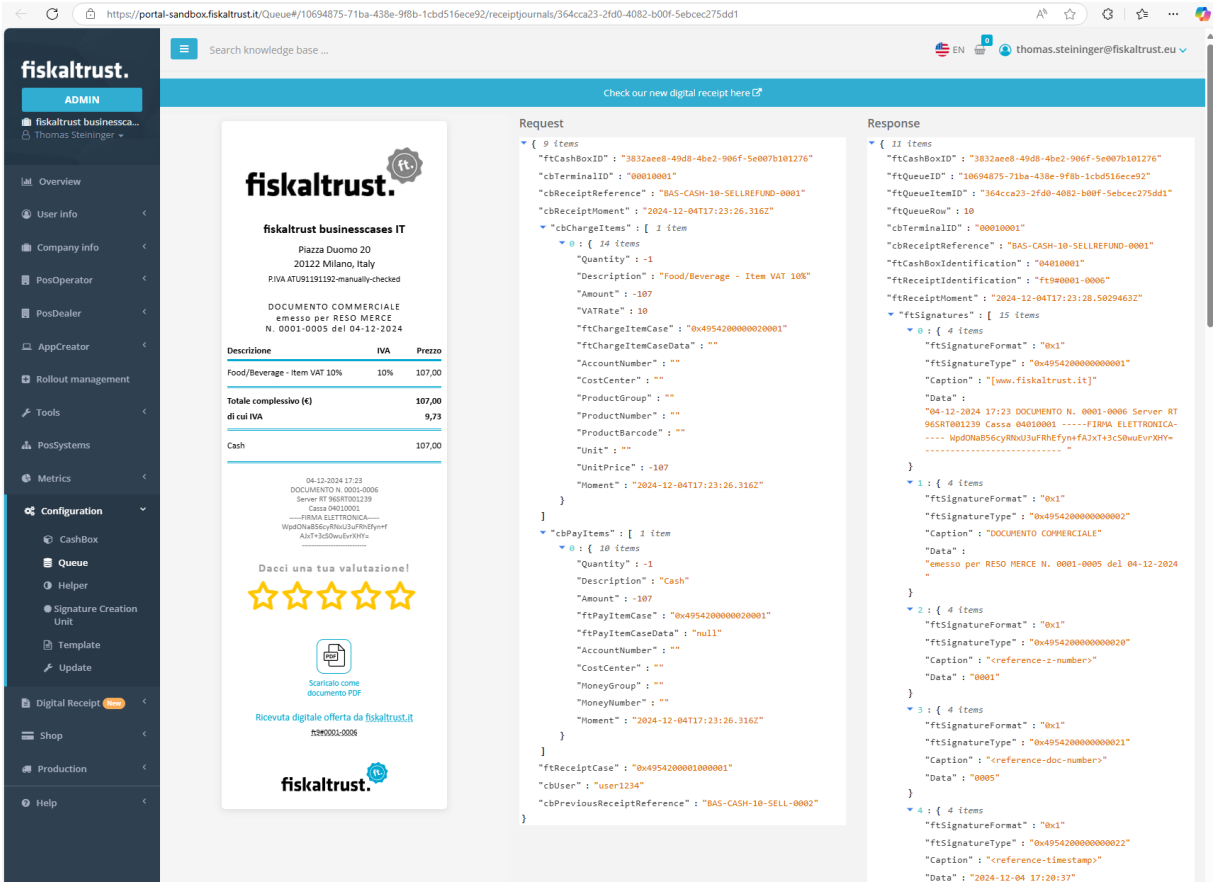
fiskaltrust.portal | Queues

Void receipt
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/843def24-0e21-4e8f-95ae-f03602b2223e



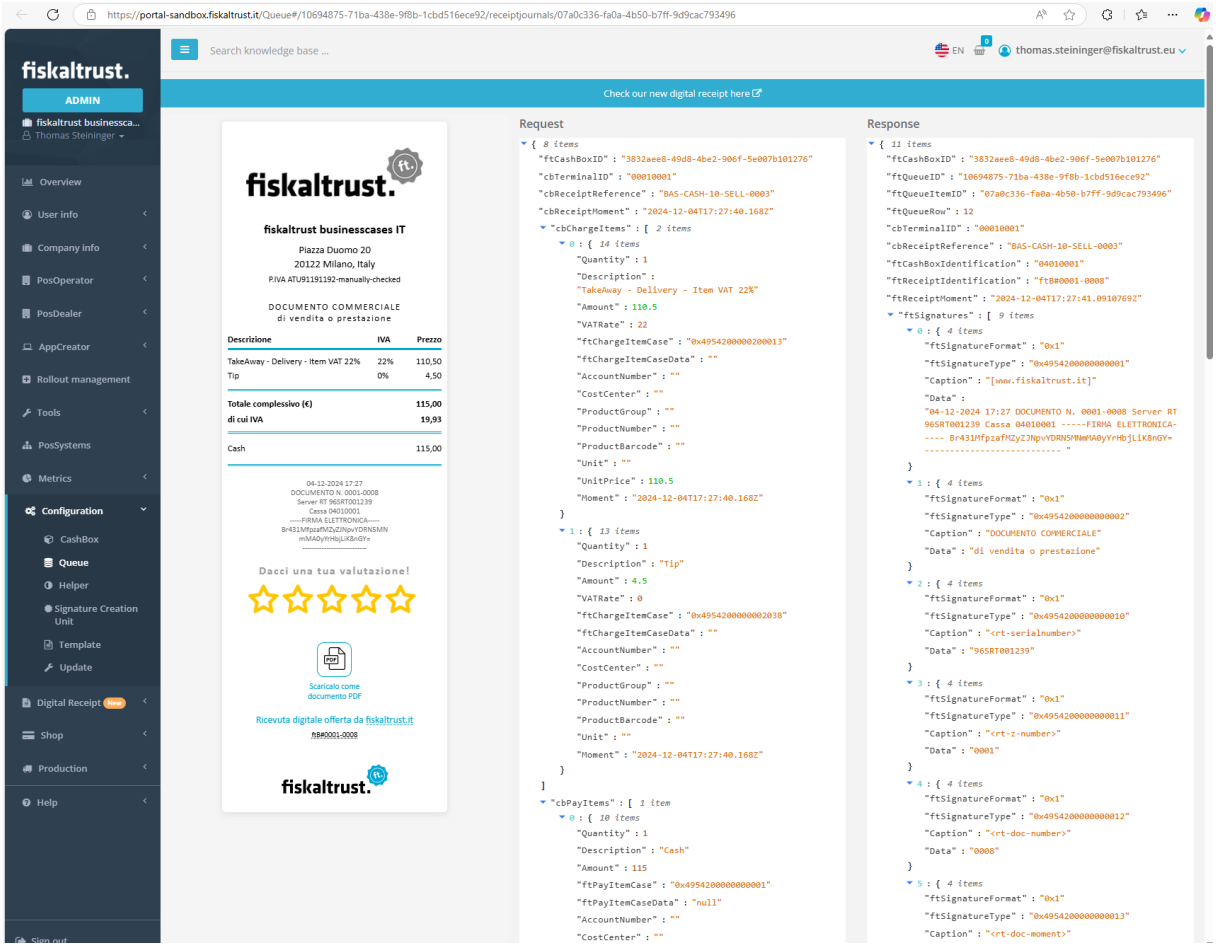
fiskaltrust.portal | Queues

Refund product
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/364cca23-2fd0-4082-b00f-5ebcec275dd1



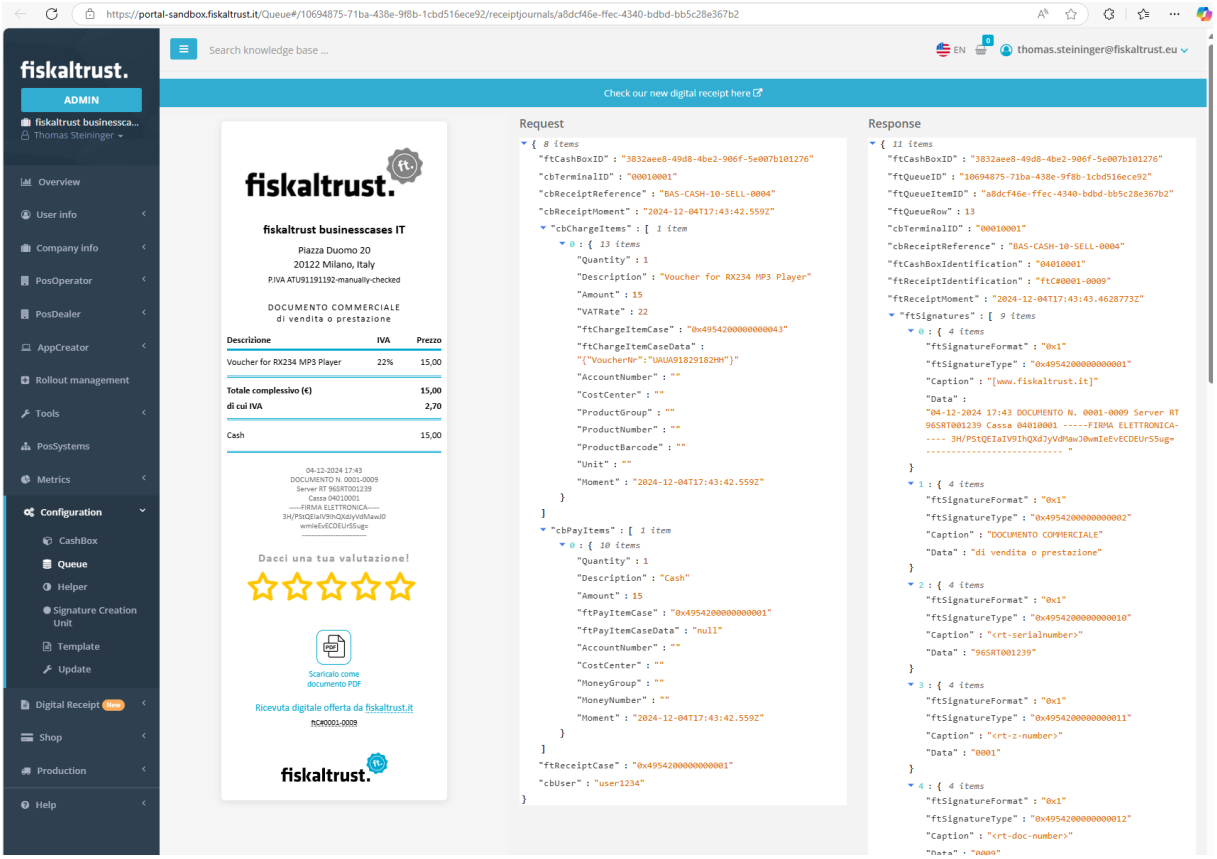
fiskaltrust.portal | Queues

Tip to owner
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/07a0c336-fa0a-4b50-b7ff-9d9cac793496



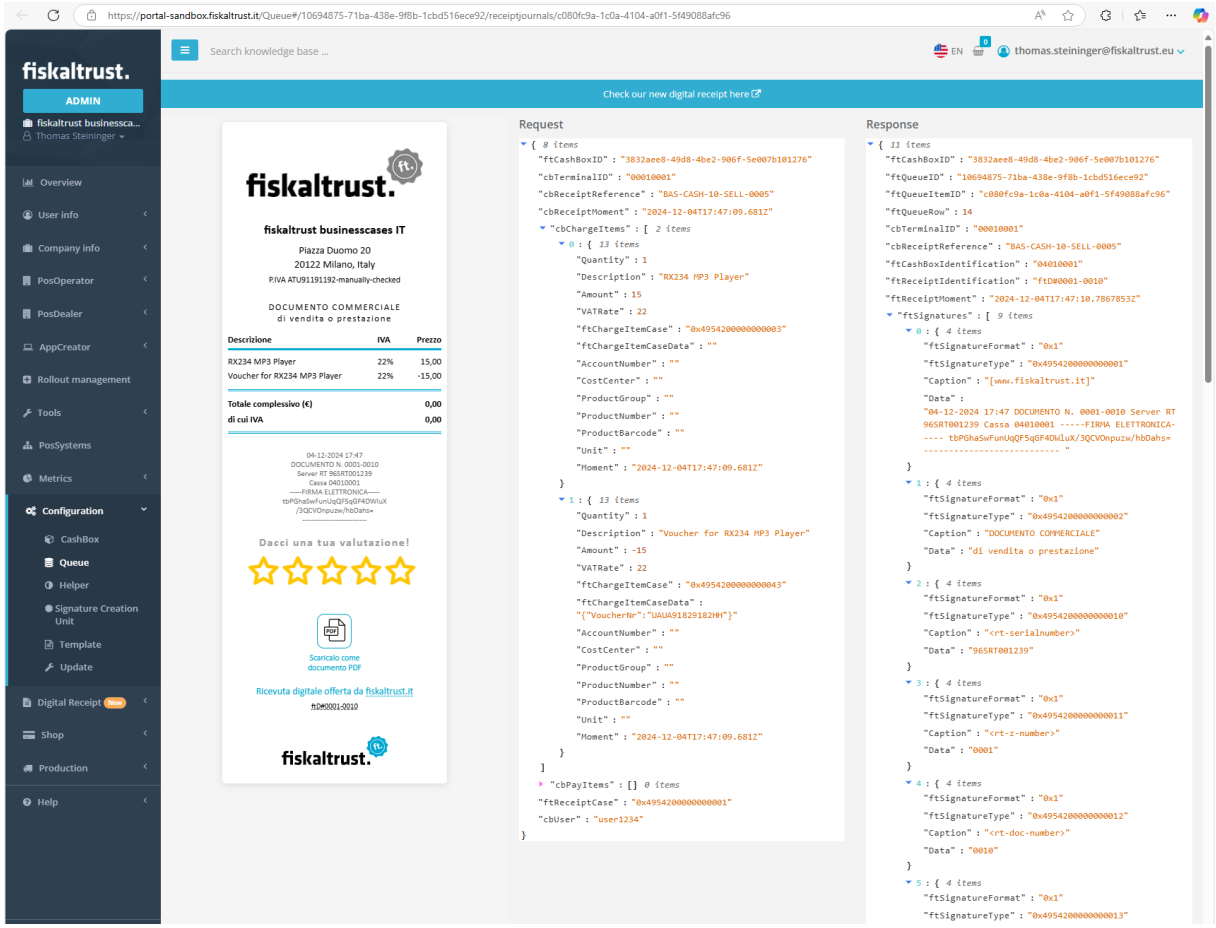
fiskaltrust.portal | Queues

Issue singlepurpose voucher
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/a8dcf46e-ffec-4340-bdbd-bb5c28e367b2



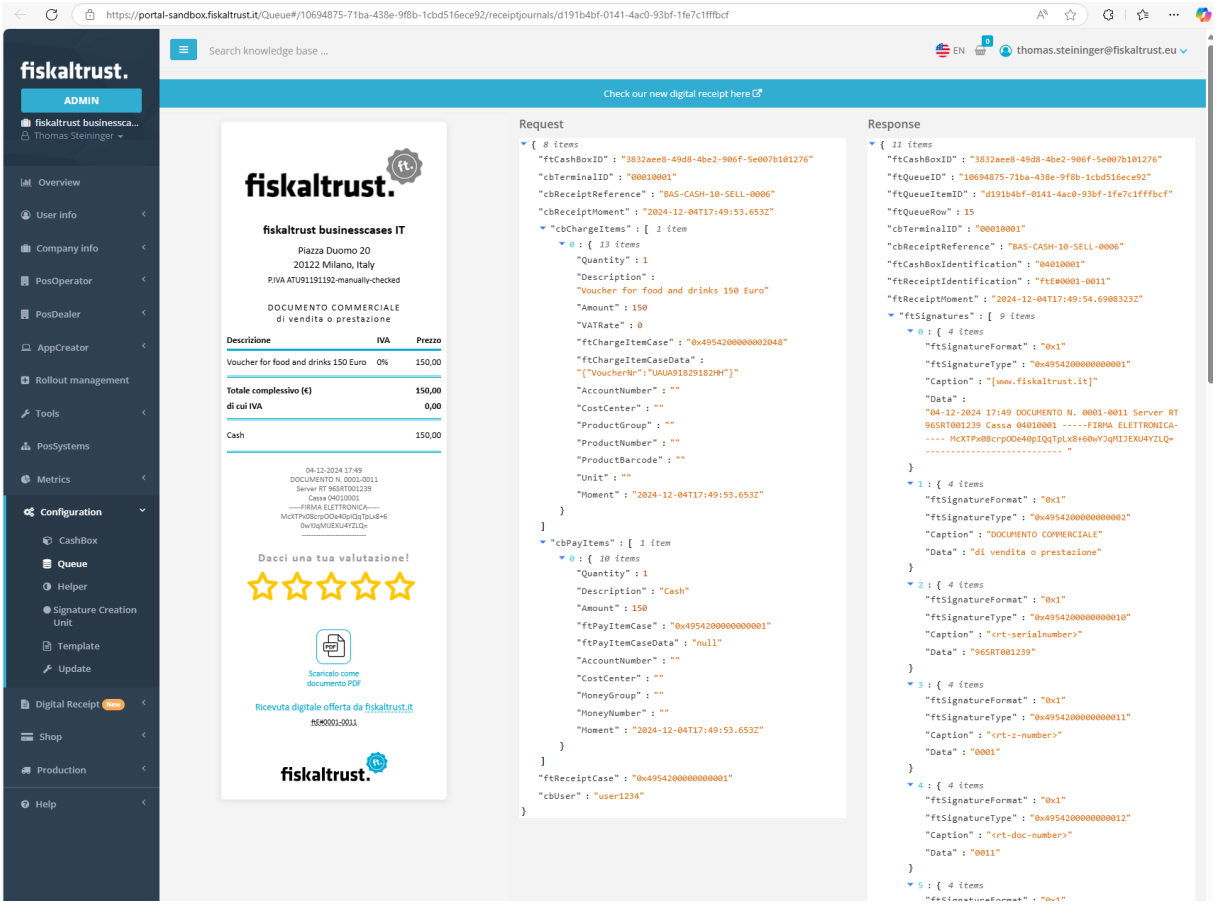
fiskaltrust.portal | Queues

Redeem singlepurpose voucher
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/c080fc9a-1c0a-4104-a0f1-5f49088afc96



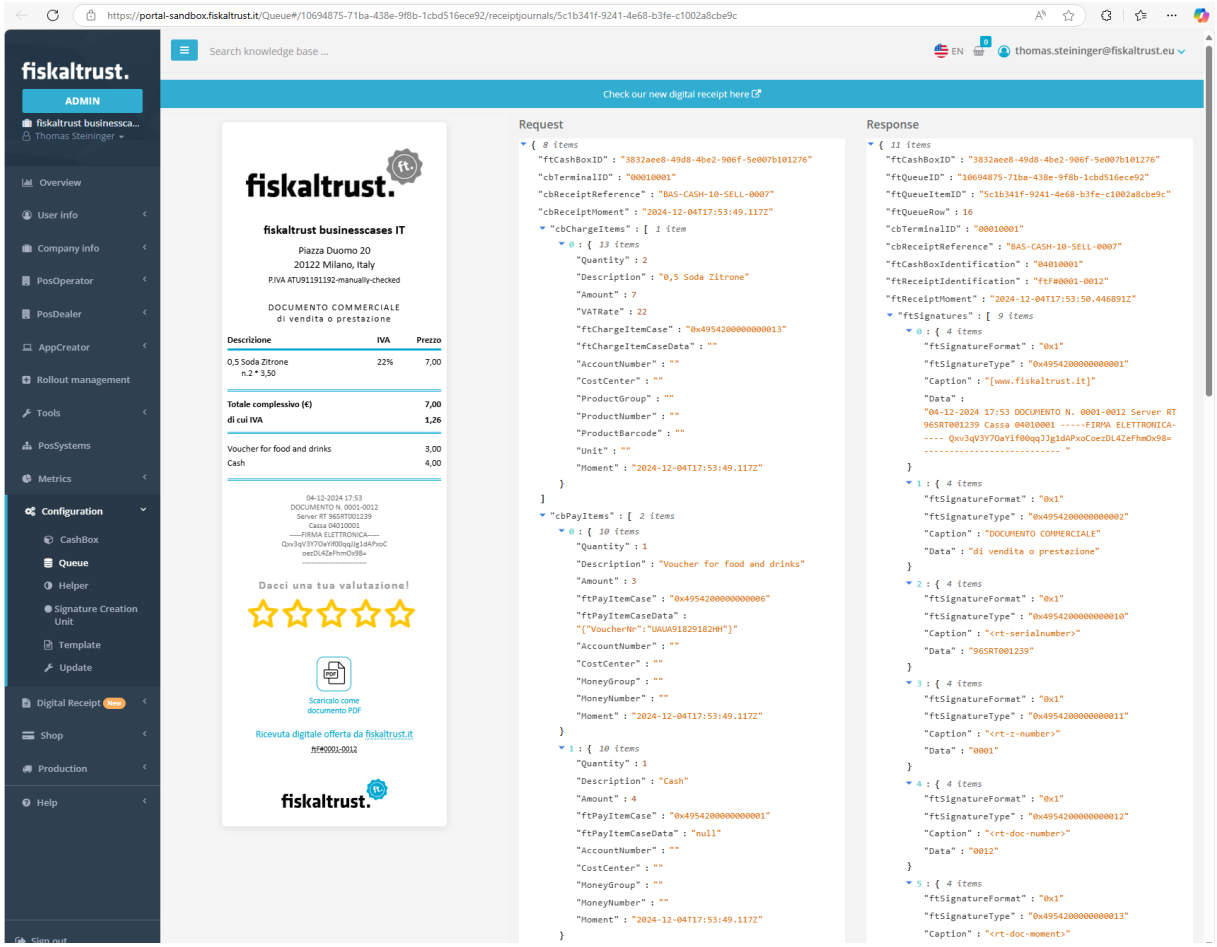
fiskaltrust.portal | Queues

Issue multipurpose voucher
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/d191b4bf-0141-4ac0-93bf-1fe7c1fffbcf



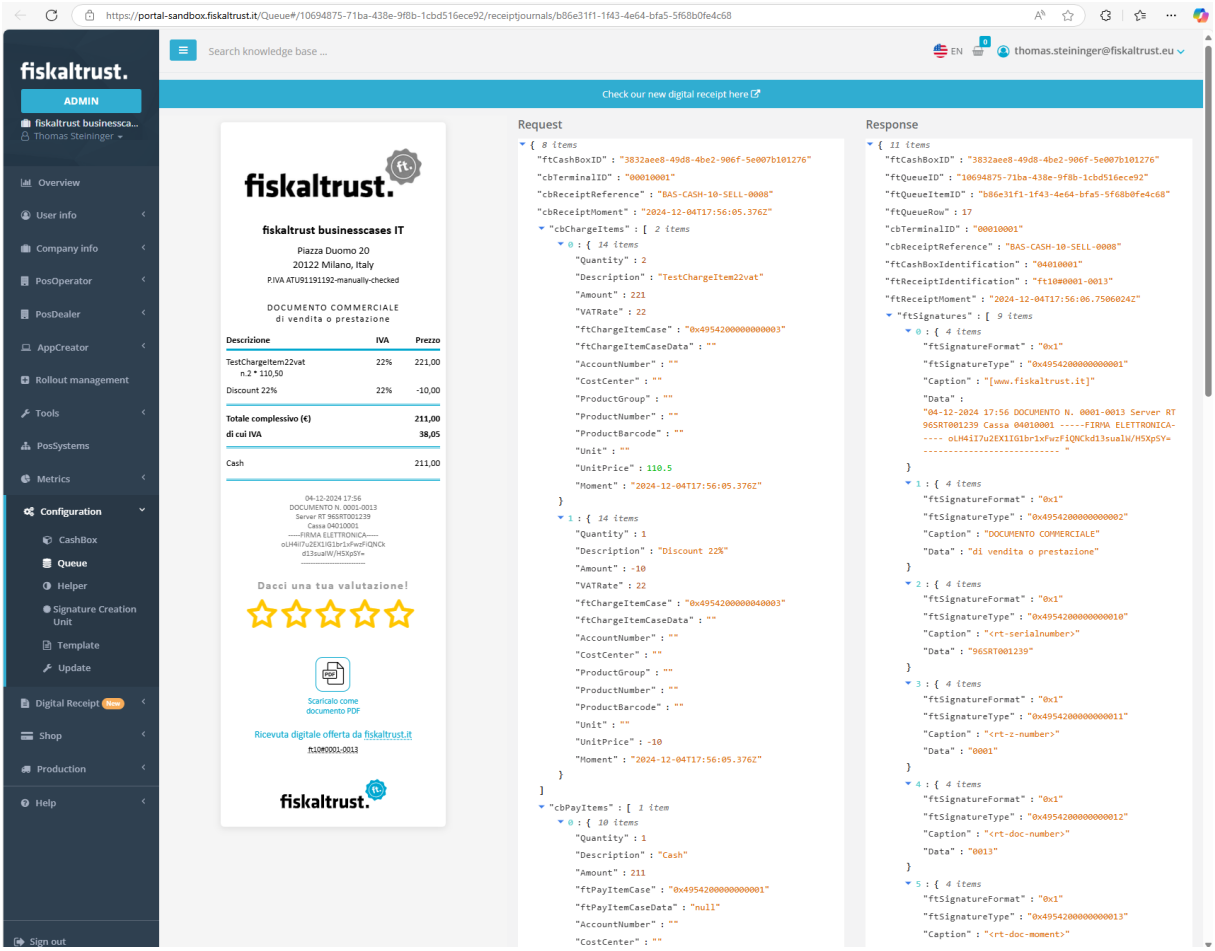
fiskaltrust.portal | Queues

Redeem multipurpose voucher
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/5c1b341f-9241-4e68-b3fe-c1002a8cbe9c



fiskaltrust.portal | Queues

Discount
https://receipts-sandbox.fiskaltrust.cloud/v0/10694875-71ba-438e-9f8b-1cbd516ece92/b86e31f1-1f43-4e64-bfa5-5f68b0fe4c68



fiskaltrust.portal | Queues